

Effect of childbirth experience on the development of early postpartum depression in mothers admitted to postnatal wards at a Tertiary care hospital in Galle, Sri Lanka: A descriptive cross-sectional study

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Abstract

Background: The maternal experience of childbirth has a significant effect on the mother-child relationship and can even lead to the development of postpartum depression (PPD). Early onset PPD is defined as depression which has an onset within four weeks after giving birth to a child. The study aimed to explore the link between childbirth experiences and the development of early PPD in mothers admitted to postnatal wards at a tertiary care center in Sri Lanka.

Methods: A descriptive cross-sectional study was conducted among 420 postnatal mothers during the first 72 hours following childbirth, admitted to the German-Sri Lanka Friendship Hospital for Women, Galle employing systematic random sampling. PPD symptoms and childbirth experience were assessed with validated Sinhala versions of the Edinburgh Postnatal Depression Scale and the Childbirth Experience Questionnaire, respectively. Data was analyzed with Chi-square and Pearson correlation.

Results: The mean age $\pm$ SD of the sample was 29.8 ± 5.438 years, 54.5% were primiparous, and 45.2% had normal vaginal deliveries. A negative childbirth experience was present among 6.9% postpartum mothers, and 11.7% had symptoms of early PPD. A significant weak positive correlation was observed between negative childbirth experience and early PPD symptoms ($r=0.245$, $p<0.001$). The presence of antenatal complications ($p=0.004$), antenatal hospital admissions ($p=0.010$), and absence of spousal support ($p=0.010$) were significantly associated with a negative childbirth experience. The development of breastfeeding problems was observed as a secondary outcome ($p<0.001$) of a negative childbirth experience.

Conclusions: The prevalence of negative childbirth experiences and the presence of early PPD symptoms in mothers were found to be low but significant. Training the staff in maternity hospitals to care for patients with empathy during the perinatal period, while providing better social support for pregnant women is recommended to ensure a positive childbirth experience.

BACKGROUND

During pregnancy, pregnant women undergo significant anatomical and physiological changes in order to nurture and accommodate the developing fetus (1). These changes begin after conception and affect every system in the body (2). Most females are susceptible to developing psychological well-being issues during pregnancy or after giving birth (3).

The unique experiences women undergo during childbirth and labor remain with them throughout their lives and can either affect positively or negatively on the mother and her child (4). The prevalence of negative birth experience ranges from 6.8 to 44%(5). Childbirth experience is affected by the support provided by the midwives, duration of labor, pain that the mother undergoes, and undergoing and emergency Caesarean section(6). The outcomes of these factors collectively contribute towards making the childbirth experience a positive or a negative one.

Mothers who had a positive childbirth experience have an increased self-esteem, self-efficacy, maternal and infant attachment, and better acceptance of the maternal role (7). Thus, the growth and development of children are less likely to be delayed or altered in mothers with a positive childbirth experience (8). Negative childbirth experience has been linked to breastfeeding problems, poor self-rated health, post-traumatic stress disorder and post-partum depression among mothers (9).

Depression and anxiety are the most widely recognized negative maternal mental health problems present during pregnancy(2). However, mothers from low-income countries are more vulnerable to depression than those from high-income countries (10).

Postpartum depression (PPD) is broadly defined as an episode of major depression occurring during the perinatal period (11). PPD can be divided into two categories: early-onset PPD and late-onset PPD. Early-onset PPD is defined as depression with an onset within four weeks after giving birth and late onset PPD is defined as depressive episodes occurring five to twelve weeks after childbirth (12).

Although it is uncommon to find depressive symptoms during the immediate period of postpartum, 2.5% of women suffer from PPD just two days after childbirth (13). The reported global pooled prevalence of PPD is 17.7%(14). Prevalence of PPD in developed countries such as the United States of America is at 13.9% whilst in Germany it is at 3.6%(15, 16). However, the prevalence of PPD in Asian countries varies considerably, ranging from 11.5% in China to between 18.23% to 32.8% in India (17–19).

Risk factors for PPD include low spousal and social support, history of depression, genetic factors, chronic illnesses, past or current abuse, diabetes, body image dissatisfaction, multiple births, negative childbirth experiences and delivering preterm and low birth weight infants (20). Further, mothers with a history of mental disorders and newborn deaths have the highest risk of developing PPD (21).

PPD results in negative outcomes in both the infant and the mother. Poor cognitive functioning, behavioral inhibition, emotional maladjustment, violent behavior, externalizing disorders, and psychiatric and mental disorders in adolescence are some of the negative outcomes among infants (22). Negative outcomes among the mothers include weight problems, alcohol and illicit drug use, social relationship problems, breastfeeding problems, and persistent depression beyond the pregnancy (22).

Mothers with negative childbirth experiences are reported to be at risk of developing PPD (23, 24). However, there is a significant gap in the literature regarding the effect of childbirth experience on the development of postpartum depression.

Sri Lanka is well recognized for its quality maternal health promotion and reduced maternal mortality rates compared to other countries in the South Asian region, whilst being considered a lower-middle-income country (25). Though Sri Lanka possesses an efficient maternal care package, maternal mental health services remain a neglected area (26).

Studies assessing the maternal mental health status during the perinatal period are sparse in the country, and reports that PPD to be present among 27.1% mothers in Sri Lanka (27). But more recent

studies reported the prevalence of PPD was 15.5% in 10 days and 7.8% at 4 weeks postpartum (28). However, a national estimate for PPD in Sri Lanka is not available yet.

The current study was the first study conducted in the country assessing childbirth experience and its association with early PPD and is one of the first few studies conducted in the entire South Asian region.

METHODS

Study Setting

A descriptive, cross-sectional study was conducted from December 2023 to May 2024 at the postnatal wards of German–Sri Lanka Friendship Hospital for Women, Galle which is the only tertiary care provider for maternal and child health in the Southern province of Sri Lanka.

Study Sample

The sample consisted of 420 mothers admitted to the postnatal wards, selected through systematic random sampling. Mothers who have delivered a baby in the past 72 hours were included in the study and mothers with a history of depression or a psychological disorder before or during pregnancy or currently suffering from a psychiatric illness, who have had a stillbirth/neonatal death during the past 72 hours, and those admitted in postnatal wards for more than 72 hours were excluded from the study.

Study instruments

Sociodemographic details and the obstetric history of the mother were obtained through a self-administered questionnaire, and delivery details were extracted from hospital records. The self-administered validated Sinhala version of the Childbirth Experience Questionnaire (CEQ) was employed to assess the experience of women in different aspects of their labor and childbirth (29). The childbirth experience questionnaire assesses childbirth experience in four areas with 22 questions, which includes own capacity, perceived safety, professional support, and participation (30).

The validated Sinhala version of the Edinburgh Postnatal Depression Scale (EPDS), which has been widely employed in the country for research and is routinely used in Sri Lanka at four to six weeks postpartum to screen for postpartum depression, was employed to assess PPD (28). The EPDS was designed to screen women for depressive symptoms during pregnancy and the postnatal period, and this scale is a valuable and efficient way of identifying patients at risk for perinatal depression (31). It is a 10-item self-report scale consisting of statements describing depressive symptoms and usually takes about five minutes to complete (27, 32).

Ethical approval

was obtained for the study from the Ethical Review Committee, Faculty of Medicine, University of Ruhuna. Informed written consent was taken from all the mothers prior to recruitment, and mothers who screened positive for symptoms of early PPD were referred to the maternal mental health clinic at the hospital.

Data analysis

Data was analyzed using Statistical Package for the Social Sciences (SPSS) software version 27. The Chi-square test was used to ascertain the associations between factors affecting childbirth experience. The relationship between childbirth experience and PPD was determined with Pearson correlation.

RESULTS

A total of 420 postpartum women in the first 72 hours of the postpartum period participated in the study. The postpartum women were in the 17 to 47-year age range with a mean age of 29.8 years (SD = 5.438).

Table 1
Sociodemographic characteristics of postpartum women admitted to postnatal wards

Socio-demographic characteristics	Category	Frequency	Percentage (%)
Maternal age	≤ 19 years	5	1.2
	20–35 years	353	84.0
	≥ 36 years	62	14.8
Education	No school education	2	0.5
	Primary education	4	1.0
	Secondary education	26	6.2
	Passed GCE O/L ¹	136	32.4
	Passed GCE A/L ²	172	41.0
	Diploma/Degree holder	80	19.0
Employment	Unemployed	257	61.2
	Employed	163	38.8
Income	Above LKR ³ 104482.0 ⁴	35	8.3
	LKR ≥ 104482.0	385	91.6
¹ General Certificate of Education Ordinary Level ² General Certificate of Education Advanced Level ³ Sri-Lankan Rupees ⁴ Average monthly income of a Sri Lankan according to the latest data published by the Central Bank of Sri Lanka			

A major percentage (84.0%) belonged to the 20-35 years age category, 1.2% accounted for pregnancies under the age of 20 years, and 14.8% accounted for pregnancies above the age of 35 years. 61.2% of the mothers were unemployed, and majority of them (91.7%) earned less than average monthly income of a Sri-Lankan. 92.4% of mothers have passed the GCE O/L examination, and 19.0% were degree/diploma holders. See Table 1

Table 2 - Obstetric characteristics of postpartum women admitted to postnatal wards

Obstetric Complications	Category	Frequency	Percentage (%)
Number of children	No children	229	54.5
	One child	140	33.3
	Two or more children	51	12.1
Mode of delivery	Normal vaginal delivery	190	45.2
	Lower segment caesarean section	230	54.8
Pregnancy-related complications	Pregnancy-induced hypertension	39	9.3
	Gestational diabetes mellitus	63	15
	Antepartum haemorrhage	11	2.6
	Preterm rupture of membranes	16	3.8
	Anaemia	28	6.7
	Other ¹	12	2.9
	No complications	251	59.8
Complications during delivery	Absent	296	70.5
	Present	124	29.5
¹ Other - Asthma, Recurrent urinary tract infections, and Dyslipidaemia			

The majority were primigravida mothers (54.5%). The number of children at home varied from one to five with most of the multiparous mothers having one child at home (33.3%). Among the mothers who had pregnancy-related complications, the majority (15%) had Gestational Diabetes Mellitus. See Table 2

Negative childbirth experience was defined as “a CEQ score less than or equal to 2.5” based on a previous practice by a prospective cohort study conducted in 2024 (33). A negative childbirth experience was present among 6.9% mothers with a mean CEQ value of 3.07 (SD = 0.38).

Positive for early postpartum depressive symptoms was defined as per the national cut-off level of nine or more in the score of the EPDS (30). Symptoms of early postpartum depression was present among 11.7% mothers with a mean EPDS score of 4.95 (SD = 3.78).

Table 3

Socio-demographic factors affecting childbirth experience in mothers in the first 72 hours postpartum

Sociodemographic factors	Negative childbirth experience		Total (%)	Significance
	Present (%)	Absent (%)		
Age				
≤ 19 and ≥ 36 years	6 (8.95)	61 (91.04)	67 (100.0)	Fisher's exact
20–35 years	23 (6.61)	325 (93.39)	348(100.0)	p = 0.441
Education level				
Up to GCE O/L ¹	8 (4.76)	160 (95.24)	168 (100.0)	χ ² = 2.000
More than GCE O/L	21 (8.33)	231 (91.66)	252 (100.0)	p = 0.157
Employment				
Employed	16(9.81)	147(90.18)	163(100.00)	χ ² = 3.512
Unemployed	13(5.05)	244(94.94)	257(100.00)	P = 0.061
Income				
≤ Mean monthly income ²	24(6.23)	361(93.76)	385 (100.00)	Fisher's Exact
>Mean monthly income ²	5(14.28)	30(85.71)	35(100.00)	P = 0.082
Availability of support from husband				
Present	23 (5.88)	368 (94.12)	391 (100.0)	Fisher's exact
Absent	6 (20.69)	23 (79.31)	29 (100.0)	p = 0.010*
¹ General Certificate of Education Ordinary Level 2 Mean monthly income of an average Sri Lankan is Sri Lankan Rupees 104482.0				

A negative childbirth experience is associated with living without a husband following discharge from the hospital: (20.69% vs 5.88%, p=0.010). See Table 3

Table 4

Obstetric factors affecting childbirth experience in mothers in the first 72 hours postpartum

Obstetric Factors	Negative childbirth experience		Total (%)	Significance
	Present (%)	Absent (%)		
Parity				
Primigravida	14 (6.11)	215 (93.89)	229 (100.0)	χ ² = 0.490
Multigravida	15 (7.85)	176 (92.15)	191 (100.0)	p = 0.484
If multiparous, number of children present				
One child	29 (7.09)	380 (92.91)	409 (100.0)	Fisher's exact
Two or more children	0 (0)	11 (100)	11 (100.0)	p = 1.000
Mode of delivery				
NVD ¹	16 (8.42)	174 (91.58)	190 (100.0)	χ ² = 1.241
LSCS ²	13 (5.65)	217 (94.35)	230 (100.0)	p = 0.265
Antenatal complications				
Present	19 (11.24)	150 (88.76)	169 (100.0)	χ ² = 8.278
Absent	10 (3.98)	241 (96.02)	251 (100.0)	p = 0.004 *
Premature Baby Unit (PBU) admissions				
Present	7 (14.29)	42 (85.71)	49 (100.0)	Fisher's exact
Absent	22 (5.93)	349 (94.07)	371(100.0)	p = 0.063
Hospital admissions in the antenatal period				
Present	9 (9.47)	86 (90.53)	95 (100.0)	Fisher's exact
Absent	20 (6.18)	304 (93.82)	324(100.0)	p = 0.010*
Problems in breastfeeding				
Present	20 (12.74)	137 (87.26)	157 (100.0)	χ ² = 13.106
Absent	9 (3.45)	252 (96.55)	261 (100.0)	p < 0.001*
Date of delivery				
Delivered on due date	1 (1.43)	69 (98.57)	70 (100.0)	Fisher's exact
Not delivered on the due date	28 (8.0)	322 (92)	350 (100.0)	p = 0.066
¹ Normal Vaginal Delivery ² Lower Segment Caesarean Section				

Obstetric Factors	Negative childbirth experience		Total (%)	Significance
	Present (%)	Absent (%)		
Parity				
History of miscarriage				
Present	6 (7.89)	70 (92.10)	76 (100.0)	χ ² = 0.132
Absent	23 (6.72)	319 (93.27)	342 (100.0)	p = 0.717
Complications during delivery				
Present	4 (3.22)	120 (96.77)	124 (100.0)	χ ² = 3.705
Absent	25 (8.44)	271 (91.55)	296 (100.0)	p = 0.054
¹ Normal Vaginal Delivery ² Lower Segment Caesarean Section				

The presence of a negative childbirth experience was associated with antenatal complications (11.24 % vs 3.98%, $p=0.004$), presence of hospital admissions in the antenatal period (9.47% vs 6.18%, $p=0.010$), and having problems with breastfeeding (12.74% vs 3.45%, $p<0.001$). See Table 4

A statistically significant weak negative correlation between the CEQ and EPDS scores ($r= 0.245$, $p=<0.001$) was observed. Thus, there is a weak positive correlation between the development of early postpartum depressive symptoms and a negative childbirth experience, where women with a negative childbirth experience are more prone to develop early postpartum depressive symptoms. Refer figure 1

DISCUSSION

Negative childbirth experience was identified to be present among 6.9% mothers in the early postpartum period in our study, which is low yet significant. The cut-off value for negative childbirth experience was considered as ≤ 2.5 (33). Even though a similar small-scale cross-sectional study in India had used the median as the cutoff value, we opted to use the cutoff as per a large-scale prospective cohort study conducted in Iran (33, 34).

The prevalence of negative childbirth experience in an Indian study was reported as 52.65%, which was significantly higher than our study, which was 6.9% (34). Though Sri Lanka and India are neighboring countries with similar socio-cultural practices, the significant difference in results could be due to variations in maternal healthcare delivery (35). However, since the studies have used different cut-off levels, a direct comparison between the two studies cannot be made.

Studies which have assessed early PPD are scarce in the literature, especially in the Asian context. The presence of early postpartum depressive symptoms among the current study population is 11.7% which is in par with the prevalence of 10.2%, reported by a retrospective study in Taiwan, in 2023(36) The low prevalence could be a result of the constant emotional support provided by the PHMs throughout the

pregnancy and during the hospitalization by providing individualized care(37) Further, social support received by the mother due to the sociocultural background in the country can play a key role as well (38).

A significant association between negative childbirth experience and the presence of antenatal complications, presence of hospital admissions in the antenatal period, and the social support that the mother would receive following childbirth, especially the support from her husband, was observed. Low awareness of antenatal complications among pregnant women can lead to undue stress and concern over the entire pregnancy as well as during childbirth (39). Therefore, health education on childbirth is crucial for expectant mothers, are well-informed and empowered, thereby reducing pregnancy-related anxiety (39).

A cross-sectional survey done in the United Kingdom in 2023 revealed that mothers with poor antenatal care are more prone to have labour complications (40). Therefore, antenatal complications, recurrent admissions, and the poor care they receive might lead to delivery complications and ultimately lead to a negative experience during delivery.

Pregnancy is a transforming period where the mother undergoes significant mental and physical changes (41). Support from the family is observed to be crucial get through such an extensive change successfully and aligns with the findings of our study as well(42). The knowledge that the mother will receive support from her husband seems to have played a key role in determining the childbirth experience(43). An analytic cross-sectional survey conducted in Iran in 2019 reported that social support provided by significant others (spouse) during pregnancy has an important effect on a woman`s childbirth experience, as was observed in the current study (44).

Though increasing age was not found as a factor affecting negative childbirth experience in our study, a Canadian study conducted in 2006 revealed an association between negative childbirth experience and older age (45).Unemployment, operative modes of birth which include unplanned Caesarean sections and instrumental vaginal births, have shown significant associations to the development of negative childbirth experience though it was not found significant through our study (9, 46).

As a secondary outcome, it was found that a negative childbirth experience leads to problems in breastfeeding in the current study. The severe mental stress caused during an unpleasant delivery can lead to problems in lactation, leading to breastfeeding problems (47). Several studies conducted in the UK have revealed that mothers who had more positive birth experiences were more likely to breastfeed their babies better than those who had more negative experiences (48, 49).

CONCLUSIONS

The prevalence of negative childbirth experience among the study population was 6.9%, which was low yet significant, and postpartum depressive symptoms in the immediate postpartum were present among 11.7% postpartum mothers, which is a concerning factor.

There is a statistically significant weak positive correlation between childbirth experience and the development of early postpartum depressive symptoms among mothers in the early postpartum period.

Factors affecting negative childbirth experience include the presence of antenatal complications, hospital admissions in the antenatal period, and living without the husband following the delivery.

The presence of breastfeeding problems was a secondary outcome of a negative childbirth experience.

In the Sri Lankan setting, knowledge regarding the progress of a normal pregnancy is provided to mothers via the antenatal sessions conducted by the MCH clinics, but individualized care on the specific complications provided by primary health care workers to mothers with antenatal complications is low. Therefore, it is recommended to increase awareness about specific health conditions by enhancing the one-to-one care given by public health midwives and physicians at the hospitals. Further, conducting special programs targeting mothers with antenatal complications with the participation of obstetricians would be beneficial during the antenatal period.

Including components on how to prepare mother's mental status for childbirth, and encouraging active participation of the partner or a family member along with the mother for the antenatal sessions is recommended.

In Sri Lanka, the presence of postpartum depressive symptoms is assessed using the EPDS only at four to six weeks postpartum. The current study revealed that one in ten mothers have early postpartum depressive symptoms within seventy-two hours postpartum. Therefore, we recommend to screen the postpartum mothers on discharge from the hospital after the delivery and referring them for early interventions. As breastfeeding problems were a secondary outcome of negative childbirth experience, it is recommended to have a focused special focus on postnatal mothers who had antenatal complications, through the lactation management units during the early postpartum period.

Abbreviations

PPD – Postpartum Depression

EPDS – Edinburgh Postnatal Depression Scale

CEQ – Childbirth Experience Questionnaire

PBU – Premature Baby Unit

GCE – General Certificate of Education

O/L – Ordinary Level

A/L – Advanced Level

SPSS – Statistical Package for Social Sciences

MCH – Maternal and Child Health

PHM – Public Health Midwife

Declarations

Ethics approval and consent to participate

Ethics approval was obtained from the Ethical Review Committee, Faculty of Medicine, University of Ruhuna on 25.09.2023 (Ref: 2023/P/074 (20.07.2023)) Written informed consent was obtained from all the participants prior to recruitment for the study. The study was conducted in accordance with the principles outlined in the Declaration of Helsinki.

Consent for publication

Not applicable

Availability of data and materials

The data analyzed to obtain the findings for this study are available from the corresponding author upon reasonable request.

Competing interests

The authors declare no competing interests.

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Authors' contributions

All authors designed the study and interpreted the data. All authors were responsible for the collection of data. T.M.K. performed the statistical calculations. All authors contributed to the drafting of the manuscript. Final compilation was done by T.M.K. and was reviewed by V.D. All authors read and approved the manuscript.

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References

1. Soma-Pillay P, Nelson-Piercy C, Tolppanen H, Mebazaa A. Physiological changes in pregnancy. *Cardiovasc J Afr* [Internet]. 2016 Mar 1 [cited 2023 May 11];27(2):89. Available from: [/pmc/articles/PMC4928162/](https://pubmed.ncbi.nlm.nih.gov/264928162/)
2. Chauhan A, Potdar J. Maternal Mental Health During Pregnancy: A Critical Review. *Cureus* [Internet]. 2022 Oct 25 [cited 2023 May 11];14(10). Available from: <https://pubmed.ncbi.nlm.nih.gov/36426343/>
3. Chauhan A, Potdar J. Maternal Mental Health During Pregnancy: A Critical Review. *Cureus*. 2022 Oct;14(10):e30656.
4. Hosseini Tabaghdehi M, Kolahdozan S, Keramat A, Shahhossein Z, Moosazadeh M, Motaghi Z. Prevalence and factors affecting the negative childbirth experiences: a systematic review. *J Matern Fetal Neonatal Med*. 2020 Nov;33(22):3849–56.
5. Hosseini Tabaghdehi M, Kolahdozan S, Keramat A, Shahhossein Z, Moosazadeh M, Motaghi Z. Prevalence and factors affecting the negative childbirth experiences: a systematic review. *J Matern Fetal Neonatal Med*. 2020 Nov;33(22):3849–56.
6. Hildingsson I, Johansson M, Karlström A, Fenwick J. Factors Associated With a Positive Birth Experience: An Exploration of Swedish Women's Experiences. *Int J Childbirth*. 2013 Jan 1;3(3):153–64.
7. Ekström A, Nissen E. A mother's feelings for her infant are strengthened by excellent breastfeeding counseling and continuity of care. *Pediatrics*. 2006 Aug;118(2):e309-14.
8. Hosseini Tabaghdehi M, Keramat A, Kolahdozan S, Shahhosseini Z, Moosazadeh M, Motaghi Z. Positive childbirth experience: A qualitative study. *Nurs Open*. 2020 Jul;7(4):1233–8.
9. Viirman F, Hess Engström A, Sjömark J, Hesselman S, Sundström Poromaa I, Ljungman L, et al. Negative childbirth experience in relation to mode of birth and events during labour: A mixed methods study. *Eur J Obstet Gynecol Reprod Biol*. 2023 Mar;282:146–54.
10. Adewuya A, Aydin N, Barnett B, Bertolote J, Cooper P, Chaterjee S, et al. Maternal mental health and child health and development in resource-constrained settings. *Hanoi*; 2007 Jun.
11. Kroska EB, Stowe ZN. Postpartum Depression: Identification and Treatment in the Clinic Setting. *Obstet Gynecol Clin North Am*. 2020 Sep;47(3):409–19.
12. Mori T, Tsuchiya KJ, Matsumoto K, Suzuki K, Mori N, Takei N. Psychosocial risk factors for postpartum depression and their relation to timing of onset: The Hamamatsu Birth Cohort (HBC) Study. *J Affect Disord*. 2011 Dec 1;135(1–3):341–6.
13. Pawar G, Wetzker C, Gjerdingen D. Prevalence of depressive symptoms in the immediate postpartum period. *J Am Board Fam Med* [Internet]. 2011 May [cited 2023 May 11];24(3):258–61. Available from: <https://pubmed.ncbi.nlm.nih.gov/21551397/>
14. Hahn-Holbrook J, Cornwell-Hinrichs T, Anaya I. Economic and Health Predictors of National Postpartum Depression Prevalence: A Systematic Review, Meta-analysis, and Meta-Regression of

- 291 Studies from 56 Countries. *Front Psychiatry*. 2017;8:248.
15. Getahun D, Oyelese Y, Peltier M, Yeh M, Chiu VY, Takhar H, et al. Trends in Postpartum Depression by Race/Ethnicity and Pre-pregnancy Body Mass Index. *Am J Obstet Gynecol*. 2023 Jan;228(1):S122–3.
 16. v Ballestrem CL, Strauss M, Kächele H. Contribution to the epidemiology of postnatal depression in Germany–implications for the utilization of treatment. *Arch Womens Ment Health*. 2005 May;8(1):29–35.
 17. Peng S, Lai X, Du Y, Meng L, Gan Y, Zhang X. Prevalence and risk factors of postpartum depression in China: A hospital-based cross-sectional study. *J Affect Disord*. 2021 Mar 1;282:1096–100.
 18. Upadhyay RP, Chowdhury R, Aslyeh Salehi, Sarkar K, Singh SK, Sinha B, et al. Postpartum depression in India: a systematic review and meta-analysis. *Bull World Health Organ*. 2017 Oct 1;95(10):706–717C.
 19. Kumari P, Basu S. Postpartum Depression and Its Determinants: A Cross-Sectional Study. *Cureus*. 2024 Nov;16(11):e74044.
 20. Agrawal I, Mehendale AM, Malhotra R. Risk Factors of Postpartum Depression. *Cureus* [Internet]. 2022 Oct 31 [cited 2023 May 11];14(10). Available from: /pmc/articles/PMC9711915/
 21. Røysted-Solås T, Hinderaker SG, Ubeseckara L, De Silva V. Mothers at risk of postpartum depression in Sri Lanka: A population-based study using a validated screening tool. *PLoS One* [Internet]. 2022 May 1 [cited 2023 May 4];17(5). Available from: /pmc/articles/PMC9132337/
 22. Slomian J, Honvo G, Emonts P, Reginster JY, Bruyère O. Consequences of maternal postpartum depression: A systematic review of maternal and infant outcomes. *Women's Health* [Internet]. 2019 [cited 2023 May 22];15. Available from: /pmc/articles/PMC6492376/
 23. Bell AF, Andersson E. The birth experience and women's postnatal depression: A systematic review. *Midwifery*. 2016 Aug;39:112–23.
 24. Rosseland LA, Reme SE, Simonsen TB, Thoresen M, Nielsen CS, Gran ME. Are labor pain and birth experience associated with persistent pain and postpartum depression? A prospective cohort study. *Scand J Pain*. 2020 Jul 28;20(3):591–602.
 25. Senanayake H, Goonewardene M, Ranatunga A, Hattotuwa R, Amarasekera S, Amarasinghe I. Achieving millennium development goals 4 and 5 in Sri Lanka. *BJOG*. 2011 Sep;118(SUPPL. 2):78–87.
 26. Hapangama A, Kuruppuarachchi KALA. Maternal mental health services in Sri Lanka: challenges and solutions. *BJPsych Int* [Internet]. 2021 Nov [cited 2023 May 11];18(4):100–2. Available from: <https://www.cambridge.org/core/journals/bjpsych-international/article/maternal-mental-health-services-in-sri-lanka-challenges-and-solutions/4FB156DA156D1D69C1B482FC8F79E2AD>
 27. Agampodi TC, Agampodi SB, Wickramasinghe WA, Adhikari AM, Chathurani HK. Post partum depression - a problem that needs urgent attention. *Ceylon Med J* [Internet]. 2011 [cited 2023 May 22];56(4):183–4. Available from: <https://pubmed.ncbi.nlm.nih.gov/22298220/>

28. Fan Q, Long Q, De Silva V, Gunarathna N, Jayathilaka U, Dabrera T, et al. Prevalence and risk factors for postpartum depression in Sri Lanka: A population-based study. *Asian J Psychiatr*. 2020 Jan 1;47:101855.
29. Dencker A, Taft C, Bergqvist L, Lilja H, Berg M. Childbirth experience questionnaire (CEQ): development and evaluation of a multidimensional instrument. *BMC Pregnancy Childbirth*. 2010 Dec 10;10:81.
30. Rowel D, Jayawardena P, Fernando N. Validation of the Sinhala translation of Edinburgh Postnatal Depression Scale. *Ceylon Med J [Internet]*. 2008 [cited 2023 May 22];53(1):10–3. Available from: <https://cmj.sljol.info/articles/10.4038/cmj.v53i1.219>
31. El-Hachem C, Rohayem J, Bou Khalil R, Richa S, Kesrouani A, Gemayel R, et al. Early identification of women at risk of postpartum depression using the Edinburgh Postnatal Depression Scale (EPDS) in a sample of Lebanese women. *BMC Psychiatry [Internet]*. 2014 Sep 7 [cited 2023 May 22];14(1). Available from: <https://pubmed.ncbi.nlm.nih.gov/25193322/>
32. Cox JL, Holden JM, Sagovsky R. Detection of postnatal depression. Development of the 10-item Edinburgh Postnatal Depression Scale. *Br J Psychiatry [Internet]*. 1987 Jun;150:782–6. Available from: <http://www.ncbi.nlm.nih.gov/pubmed/3651732>
33. Nahae J, Rezaie M, Abdoli E, Mirghafourvand M, Ghanbari-Homaie S, Jafarzadeh M. Association of childbirth experience with long-term psychological outcomes: a prospective cohort study. *Reproductive Health* . 2024 Dec 1;21(1).
34. Shiva L, Desai G, Satyanarayana VA, Venkataram P, Chandra PS. Negative Childbirth Experience and Post-traumatic Stress Disorder - A Study Among Postpartum Women in South India. *Front Psychiatry*. 2021;12:640014.
35. Wyatt S, Ostbye T, De Silva V, Long Q. Antenatal depression in Sri Lanka: a qualitative study of public health midwives' views and practices. *Reprod Health*. 2022 Jan 28;19(1):23.
36. Tsai JM, Tsai LY, Tsay SL, Chen YH. The prevalence and risk factors of postpartum depression among women during the early postpartum period: a retrospective secondary data analysis. *Taiwan J Obstet Gynecol*. 2023 May;62(3):406–11.
37. Wyatt S, Ostbye T, De Silva V, Long Q. Antenatal depression in Sri Lanka: a qualitative study of public health midwives' views and practices. *Reprod Health*. 2022 Dec 28;19(1):23.
38. Dimani HA, Hapuarachchi T. Risk Factors of Postpartum Depression in Sri Lanka (Special Reference in Western Province) ISRG PUBLISHERS Risk Factors of Postpartum Depression in Sri Lanka (Special Reference in Western Province). 2024;
39. Alnasheet AM, Abdulaal NA, Kamal N. The Prevalence and Risk Factors of Pregnancy-Related Anxiety in Bahrain. *Cureus*. 2024 Apr 1;
40. Cross H, Krahé C, Spiby H, Slade P. Do antenatal preparation and obstetric complications and procedures interact to affect birth experience and postnatal mental health? *BMC Pregnancy Childbirth*. 2023 Dec 1;23(1).

41. Bjelica A, Cetkovic N, Trninic-Pjevic A, Mladenovic-Segedi L. The phenomenon of pregnancy – a psychological view. *Ginekol Pol.* 2018 Feb 28;89(2):102–6.
42. Bedaso A, Adams J, Peng W, Sibbritt D. The relationship between social support and mental health problems during pregnancy: a systematic review and meta-analysis. *Reprod Health.* 2021 Jul 28;18(1):162.
43. Schmitt N, Striebich S, Meyer G, Berg A, Ayerle GM. The partner’s experiences of childbirth in countries with a highly developed clinical setting: a scoping review. *BMC Pregnancy Childbirth.* 2022 Oct 3;22(1):742.
44. Zamani P, Ziaie T, Lakeh NM, Leili EK. The correlation between perceived social support and childbirth experience in pregnant women. *Midwifery.* 2019 Aug;75:146–51.
45. Smarandache A, Kim THM, Bohr Y, Tamim H. Predictors of a negative labour and birth experience based on a national survey of Canadian women. *BMC Pregnancy Childbirth.* 2016 May 18;16(1):114.
46. Vedeler C, Eri TS, Nilsen RM, Blix E, Downe S, van der Wel KA, et al. Women’s negative childbirth experiences and socioeconomic factors: Results from the Babies Born Better survey. *Sexual & Reproductive Healthcare.* 2023 Jun 1;36:100850.
47. Dimitraki M, Tsikouras P, Manav B, Gioka T, Koutlaki N, Zervoudis S, et al. Evaluation of the effect of natural and emotional stress of labor on lactation and breast-feeding. *Arch Gynecol Obstet.* 2016 Feb;293(2):317–28.
48. Davis AMB, Sclafani V. Birth Experiences, Breastfeeding, and the Mother-Child Relationship: Evidence from a Large Sample of Mothers. *Canadian Journal of Nursing Research.* 2022 Dec 7;54(4):518–29.
49. Power C, Williams C, Brown A. Does a mother’s childbirth experience influence her perceptions of her baby’s behaviour? A qualitative interview study. *PLoS One.* 2023 Apr 6;18(4):e0284183.

Figures

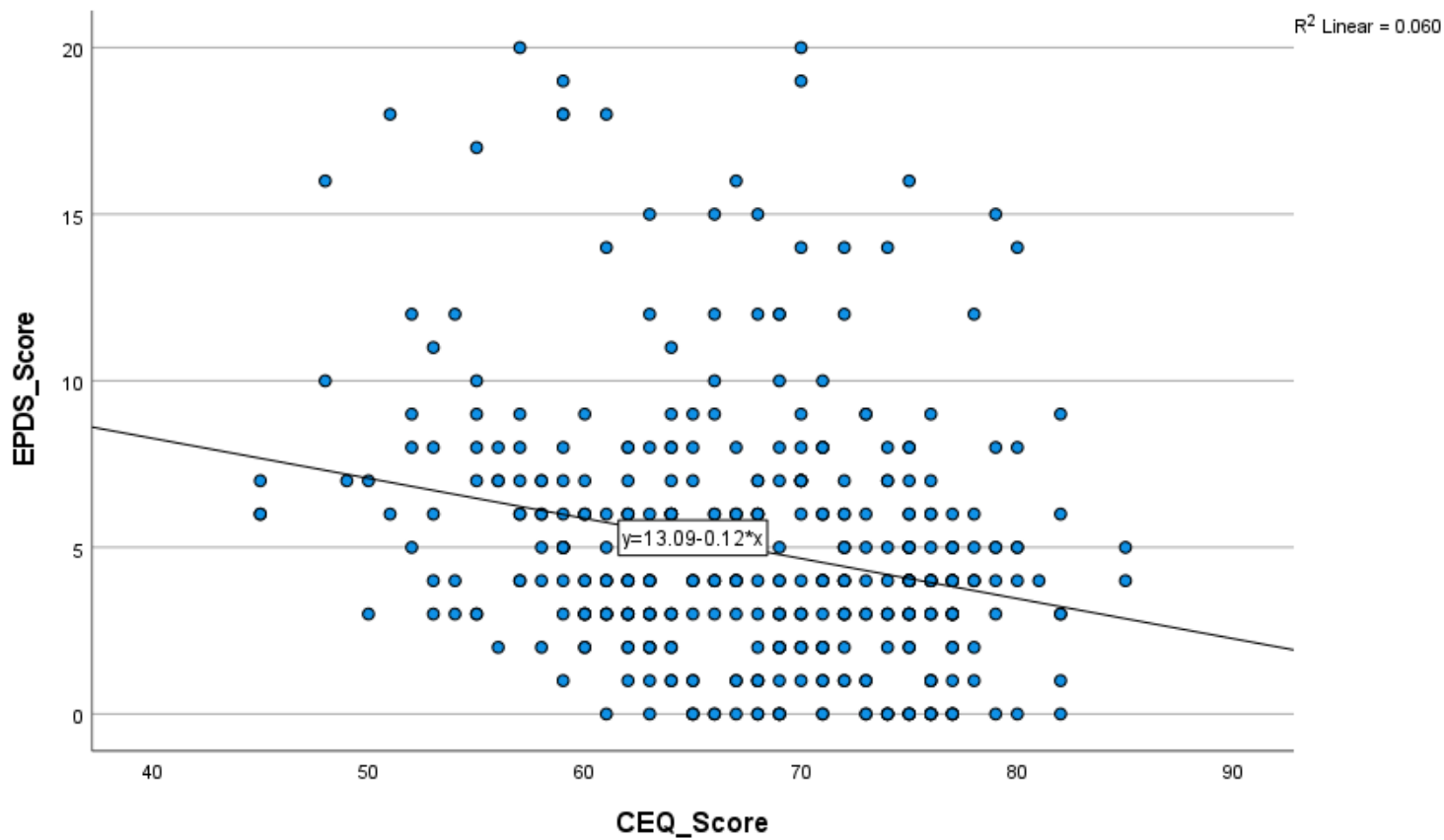


Figure 1

Correlation between EPDS score and the CEQ scores